Europe's Ancient Forests Were Nothing Like We Imagined

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Picture the wilderness of central Europe 100,000 years ago, or even a million years ago. You might imagine an endless expanse of towering trees, with thick **canopies** blocking sunlight from reaching the ground. But according to a major new study, that familiar image of Europe's ancient forests is probably misleading.

Instead, much of the continent was a vibrant landscape of scattered trees, open woodlands, grassy clearings, and colorful wildflower meadows. Birds and butterflies flourished in these sunlit habitats, while enormous grazing animals helped prevent trees and shrubs from taking over.

Researchers led by Aarhus University have found that this mixture of habitats dominated Europe's landscapes for more than 20 million years. Far from being the continent's original natural condition, the dense, dark forests common in many parts of Europe today appear to be a relatively recent development in its ecological history.

Published in the journal Biological Conservation, the findings challenge a longstanding assumption about what Europe's wilderness looked like before widespread human intervention. They also raise important questions about modern **reforestation** programs, which often prioritize planting dense forests to combat climate change and protect **biodiversity**.

"The study shows that current **reforestation** practices are on the wrong track both here in Denmark, where subsidies are only granted for planting dense forests, and elsewhere in Europe. This will not only be harmful for **biodiversity**; it will be in direct contradiction to the type of ecosystems that Europe's species have evolved in over millions of years," says Professor Jens-Christian Svenning of Aarhus University, senior author of the study.

For decades, conservation strategies have been influenced by the "closed-forest **paradigm**." This is the belief that dense forests with nearly continuous tree **canopies** were Europe's natural landscape before humans began clearing land for agriculture and settlements. Under this view, restoring nature often means allowing trees to reclaim open areas or planting forests where they no longer exist.

The new research suggests that this approach may overlook a crucial part of Europe's ecological history. Rather than being uniformly covered by dense forests, the continent was characterized by changing combinations of grasslands, shrubs, and woodlands with different amounts of tree cover. These landscapes supported a rich variety of species adapted to both wooded and open environments. Large plant-eating animals also played a major role in maintaining this diversity by grazing, browsing, and disturbing vegetation.

To reconstruct Europe's ancient environments, the researchers examined an unusually extensive collection of evidence spanning millions of years. Previous studies have generally concentrated on more recent periods. This investigation extended much further into the past, examining paleoecological records from the **Miocene epoch**, which began around 23 million

years ago, through the pre-industrial era. **Paleoecology** is the study of ancient ecosystems using clues preserved in fossils, sediments, and other natural records. Because no single type of evidence can provide a complete picture of the distant past, the team combined several independent methods.

Word Treasure

canopies -- The leafy tops of trees that spread out and can block sunlight from the ground.

reforestation -- Planting trees again in places where forests were lost.

biodiversity -- The wide variety of living things in a place.

paradigm -- A widely shared idea or way of thinking about something.

paleoecology -- The study of ancient living things and their homes, using fossils and other old clues.

Miocene epoch -- A long period in Earth's past that began around 23 million years ago.

Background you need

Forests are not all alike: some are dense, with tree tops joining into a roof that shades the ground, while others are open woodlands where sunlight reaches grass and flowers below.

Large grazing animals shape such landscapes by eating young trees and shrubs, which stops the land from closing up into thick forest.

Reforestation means planting trees where forest has been lost, and it is often done to store carbon and give wildlife a home.

To learn what a landscape looked like long ago, scientists study ancient clues such as fossils and layers of soil and sediment, a field called paleoecology.

Break it down

LEAD: The familiar image of ancient Europe as endless dense forest is probably wrong -- it was mostly open, sunny landscapes.

KEY EVENT: Aarhus University researchers found open woodlands, clearings, and meadows dominated Europe for more than 20 million years.

+- WHO: Researchers led by Aarhus University, including senior author Professor Jens-Christian Svenning.

+- WHAT: Scattered trees, grassy clearings, and wildflower meadows, kept open by large grazing animals.

+- WHEN: Over more than 20 million years, from the Miocene epoch through the pre-industrial era.

+- HOW: By combining several independent methods and paleoecological records from fossils and sediments.

+- WHY IT MATTERS: Dense forests today look recent, and planting only dense forests may harm biodiversity.

EVIDENCE: Paleoecological records spanning millions of years, including the Miocene epoch, which began around 23 million years ago.

OPEN QUESTION: How reforestation programs should change to match the open landscapes Europe's species evolved in.

Why it matters

If Europe's wildlife evolved in open, sunlit landscapes, planting only dense forests could harm the very species conservation aims to protect, so the study pushes tree-planting programs to rethink their approach.

Test what you remember

1. What did ancient Europe look like, according to the new study?

- (A) Only tall mountains and bare rock
- (B) Open woodlands, grassy clearings, and wildflower meadows
- (C) Endless dense, dark forest
- (D) A frozen desert with no plants

2. How long did this mixture of open habitats dominate Europe's landscapes?

- (A) About 1,000 years
- (B) About 100,000 years
- (C) About 200 years
- (D) More than 20 million years

3. What is the "closed-forest paradigm"?

- (A) The idea that forests should never be studied
- (B) A plan to plant trees on every farm in Denmark
- (C) A rule that forbids cutting down any tree in Europe
- (D) The belief that dense forests were Europe's natural landscape before humans cleared land

4. In which journal was the study published?

- (A) European Farming Weekly
- (B) The Journal of Space Travel
- (C) Biological Conservation
- (D) Nature Medicine

5. What did the researchers use to reconstruct Europe's ancient environments?

- (A) Paleoecological records from the Miocene epoch through the pre-industrial era
- (B) Satellite photos of modern forests
- (C) Tree rings from a single forest
- (D) Interviews with farmers

6. According to Professor Svenning, what is wrong with current reforestation practices?

- (A) They ignore the opinions of farmers
- (B) They plant dense forests, which harms biodiversity and contradicts the ecosystems species evolved in
- (C) They only plant trees in Denmark
- (D) They plant too few trees of any kind

Answer key (try the quiz first!): 1: B 2: D 3: D 4: C 5: A 6: B

Questions to think about

1. Professor Jens-Christian Svenning, senior author of the study: Current reforestation practices are on the wrong track, because subsidies in Denmark and elsewhere reward only dense forests, which harms biodiversity and contradicts the ecosystems Europe's species evolved in over millions of years.

MY THOUGHTS (at least 20 words):